



OPEN Mechanical and microscale characterization of foamed concrete with Tianqi aluminosilicate binder

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Persistent research work has aided the development of supplementary cementitious materials, contributing to both sustainable development and mitigating environmental impacts. This study utilized Tianqi aluminosilicate (TAS) as partial replacement for Ordinary Portland Cement (OPC) in foamed concrete (FC) mix. The mechanical, transport, and microstructural characteristics of the concrete were explored. The formed concrete mixes were developed by varying TAS from 0 to 40%, in steps of 10% for OPC to identify optimal performance. The mechanical characteristics (compressive, flexural, splitting tensile, and elastic modulus) improved by 18–25% over control and peaked at 20% TAS. At 20% TAS substitution, the transport properties (water absorption and permeability) improved significantly, which corresponds to 35% and 28%, reduction, respectively. SEM analysis revealed that TAS refined pore structure, yielding a denser matrix with homogeneous hydration product distribution. The result revealed foam stability and uniformity in mixes containing TAS, and an improvement in mechanical and durability of the concrete. Filler effect and pozzolanic activities of TAS were identified as two key factors responsible for the observed results. There was pore refinement improved secondary hydration in the concrete matrix. The results show that 20% TAS substitution improves strength and durability while lowering OPC use and striking the ideal performance balance. From the results, TAS proved to be a sustainable supplementary cementitious material aiding the durability of the mixes. This work advances eco-friendly construction practices by demonstrating TAS's viability in high-performance FC applications.

Keywords Tianqi aluminosilicate, Foamed concrete, Mechanical properties, Transport properties, Sustainable construction

Over the years, there has been increasing popularity in the use of foamed concrete (FC), owing to its low density, improved sound insulation, and thermal properties, ability to withstand fire and earthquakes^{1–3}. Evidence has demonstrated that these advancements result in a reduction of cost of construction and duration through lowering the requirement for formwork and minimising on-site workforce^{4–6}. FC is created by randomly incorporating air voids into the cementitious matrix, resulting in a reduced density ranging from 400 to 1750 kg/m³⁷. The presence of foam in the base mix is ascribed to a chemical response, leading to the formation of pores in the mortar slurry^{8,9}. This decreases the weight of the concrete. The increased porosity significantly reduces the overall weight of the concrete¹⁰.

Supplementary Cementitious Materials (SCMs) are recognised in the construction industry for their ability to enhance the performance, durability, and sustainability of concrete and geopolymer composites. SCMs, which include industrial by-products, are employed to replace or enhance conventional Portland cement in

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concrete, offering advantages such as reduced carbon emissions, improved workability, and increased long-term strength. Fly ash and Ground Granulated Blast-furnace Slag (GGBFS) are among the most common supplemental cementitious materials, both of which have been extensively studied and included into concrete and geopolymer systems worldwide. Fly ash is a fine, powdery material produced from the combustion of coal in power production plants. This process often produces a by-product considered waste material⁸. Fly ash has become a prominent additional cementitious material in concrete production due to its ability to enhance workability, reduce heat of hydration, and improve concrete durability. Fly ash improves the long-term strength of concrete by facilitating a pozzolanic reaction with calcium hydroxide, leading to the creation of extra cementitious compounds. Furthermore, fly ash significantly reduces the probability of alkali-silica reactions (ASR), which can cause cracking in concrete exposed to certain aggregate types. Furthermore, its use as a supplementary cementitious material reduces the environmental impact of concrete production by lessening dependence on virgin cement, so advancing sustainability in the built environment. GGBFS is a widely utilised additional cementitious material produced by the rapid cooling of molten iron slag, a by-product of iron and steel manufacturing⁹. The rapid cooling process transforms the slag into a granular form, which, when finely ground, can function as a supplementary material in concrete. GGBFS enhances the mechanical properties of concrete by increasing its compressive strength, particularly at later ages, and improving its resistance to sulphate attacks and chloride-induced corrosion. The incorporation of GGBFS improves the impermeability of concrete, making it especially beneficial in adverse environments such as sea or industrial zones. Furthermore, the use of GGBFS reduces the overall carbon footprint of concrete production by lowering the demand for clinker, the most energy-intensive component of cement.

The incorporation of these traditional SCMs into geopolymer composites has been thoroughly documented in the literature. Recent research by Deng et al.¹⁰ in the study of PVA-fibre-reinforced slag- and fly ash-blended geopolymer composites examines the impact of fly ash and GGBFS on the mechanical properties and microstructure of geopolymer composites. The study highlights the significance of these materials in improving compressive strength, flexural strength, and fracture toughness, as well as their exceptional resistance to severe chemical environments. The study emphasises the synergistic advantages of integrating PVA fibres with slag and fly ash, resulting in a composite material with superior mechanical properties and enhanced microstructure. The emergence of advanced SCMs such as Tianqi Aluminosilicate (TAS) offers a great opportunity to enhance the sustainability and performance of concrete materials. TAS, an innovative aluminosilicate material, exhibits advantageous characteristics for its reactivity and ability to form geopolymer networks. When used with concrete, it can improve the material's mechanical properties while concurrently fostering a more sustainable and environmentally friendly construction alternative. The examination of TAS and its compatibility with traditional SCMs like fly ash and GGBFS may enable the development of high-performance, low-carbon concrete materials that align with the growing demand for sustainable construction practices. By integrating novel SCMs such as TAS with traditional materials, scholars and industry professionals can enhance the performance of concrete and geopolymer composites while addressing pressing environmental challenges. As the construction industry progresses towards eco-friendly solutions, the exploration of innovative materials such as TAS will undoubtedly be crucial in shaping the future of sustainable building materials.

Exploring the use of Tianqi aluminosilicate (TAS) as a substitute for cement in FC opens promising opportunities for improving its mechanical, transport, and microstructural properties. Experimental studies have shown TAS to have the potential to improve the mechanical properties of concrete. For instance, study¹¹ showed that TAS possessed the capacity to improve compressive strength of concrete with high particle packing and decreased porosity, and up to 43% improvement in bending strength that occurred when the TAS content in TAS-alumina composites was increased from 5 to 15 mass%. Another study revealed enhanced compressive strength in FC incorporating TAS. FC with a density of 1600 kg/m³ demonstrated increases in compressive strength of up to 8–10 MPa, demonstrating the potential of TAS to achieve high compressive strengths in FC¹².

Flexural strength of concrete is also improved with TAS addition. Studies on mullite-TAS composites demonstrate that adding TAS reduces the thermal expansion coefficient while greatly enhancing flexural strength. As such, the improved performance is expected to fit FC development, thus providing increased resistance to bending stresses essential for structural applications. There was a significant increase in the flexural strength of composites containing 20% of TAS, which showed the material's influence on concrete¹³. Study¹⁴ revealed up to 156 MPa flexural strength value by adding TAS to borosilicate glass composite. A related study¹⁵ showed that TAS improved both compressive and flexural strengths, making it an ideal reinforcement for FC where mechanical performance is critical¹⁵. Ibrahim et al.¹⁶ observed that polyolefin fibres slightly enhanced compressive strength by 4.3% and flexural strength by 9.3% in FC, demonstrating the noticeable impact of fibre reinforcement when combined with TAS. Moreover, TAS enhances the hardness and toughness of materials, which are crucial for resistance to wear and fracture. The use of TAS in aluminium titanate ceramics improved the hardness and toughness of the matrix, as evident in the dense and grain interlocking in the structures. These improvements should enhance FC by giving it better resistance to impact and mechanical performance. A factor like porosity of the matrix affects the mechanical properties materials significantly. TAS significantly reduces porosity by filling voids and enhancing sintering behaviour, resulting in denser materials with greater mechanical strength and improved durability¹⁷. In FC, reduced porosity can enhance both compressive and tensile strengths, making it more robust and reliable for construction applications.

In addition, TAS improves the thermal stability and mechanical properties of concrete. TAS is essential for materials subjected to heat cycles because it preserves structural integrity at high temperatures. By doing this, FC's service life is increased, and its mechanical qualities are guaranteed to stay constant in a variety of environmental circumstances¹¹. By reducing porosity and promoting the formation of strong, interlocking grain structures, TAS ensures that FC can reliably perform under various mechanical loads and environmental conditions. These improvements highlight the ability of TAS to support the advancement of long-lasting construction materials

with outstanding reliability. Transport properties of FC (permeability, water absorption, and sorptivity) are important for determining durability and evaluating various environmental features. Adding TAS significantly enhances these properties. Studies have demonstrated that FC with a density of 1500 kg/m^3 , combined with polycarboxylate superplasticizers (SP), shows improved water absorption and sorptivity. The material's overall durability is improved by reducing water absorption to about 2.5% by volume, which is achieved by better pore distribution and smaller pore diameters at the ideal SP concentration of 1.35%¹⁸. Viscosity and workability are two crucial rheological characteristics for the processing and use of FC. Incorporating TAS and superplasticizers notably improves these properties. Research indicates that adding polycarboxylate SP to FC enhances its flowability and stability. For example, increasing the SP content up to 1.35% improves the spreadability and stability of the FC mix, resulting in better mechanical performance and smaller, more uniformly distributed pores¹⁸. Additionally, activated foaming agents treated with microwave and ultrasonic methods have been shown to decrease pore size, thereby improving compressive strength and the stability of the foam structure¹⁹.

The presence of TAS has a substantial impact on the microstructural characteristics of FC, including the arrangement of pore sizes, phase structure, and density. Various studies have shown that TAS addition in concrete enhances the microstructure's fineness, making it more homogeneous. TAS aids mullite composites reduces porosity and enhances microstructural integrity, resulting in improved mechanical properties. Specifically, TAS in mullite composites lowers the densification temperature and increases flexural strength due to the formation of a dense, interlocking grain structure¹³. Moreover, microstructural analysis of FC with varying densities ($450\text{--}750 \text{ kg/m}^3$) and silica fume content has shown that replacing cement with silica fume results in higher compressive strengths and more durable microstructures. SEM and XRD experiments demonstrate that silica fume undergoes a reaction with free calcium hydroxide in hydrated cement, resulting in the formation of calcium-silicate-hydrate (C-S-H). This compound is characterized by increased stiffness and durability compared to calcium hydroxide²⁰.

Plenty of research has been done on TAS in dense concrete and composite materials, but not as much has been done on its use in FC. The main goal of research on TAS in dense concrete is to make its mechanical attributes better, for instance its compressive strength, durability, microstructural properties. However, FC systems include a lot of voids, which changes the matrix's mechanical characteristics and how it behaves during hydration and cure⁴. There isn't much extensive research on how TAS works in these kinds of porous systems, which makes it hard to comprehend how this material interacts with structures that have air in them.

In FC systems, having macro pores in the mix changes how quickly it hydrates and how it flows. Adding TAS may change the rate at which water is absorbed differently than adding regular dense concrete. In particular, not enough research has been done on how TAS interacts with the air bubbles in FC, which make it light¹¹. Because TAS is less fluid, less workable, and less porous, it could behave very differently in FC blends than it does in more solid matrices. Because of this, it is very important that study be done to find out how TAS changes the fresh and hardened properties of FC, with a focus on how it interacts with the air-entrained matrix. One of the biggest problems with adding TAS to FC systems is that air-entrained concrete has a distinctive way of flowing. Adding air bubbles to the mix during the foaming process can modify the material's consistency and flow qualities in a big way. It's important for TAS and these air bubbles to interact because the solid particles in TAS could change the stability of the foam, which could cause the air cells to collapse or the foam structure to change.

Also, FC is less hydrated than dense concrete because it is porous⁸. This could make it harder for TAS to perform, such improve durability or thermal characteristics. TAS could also change the structure of the air voids, which are very important for FC's ability to keep heat in and block sound. It is important to know how TAS works with the air-entrained matrix and whether it could make the foam stable to move forward with its use in FC. It is also important to investigate the complex interactions between the silica-alumina parts of TAS and the hydration products in the FC matrix¹⁶. Changes in the pore structure or the production of more gel phases could modify the mechanical and thermal properties of FC. More research is needed to find the right balance between strengthening effects and keeping FC's lightweight qualities, especially when using varied amounts of TAS in the mix.

According to the review mentioned above, the addition of TAS to cement-based materials greatly improve their fresh and harden states properties. These improvements result in enhanced durability, workability, and mechanical performance, rendering cement-based materials more resilient and adaptable for a wide range of building applications. Therefore, this study investigated the possible application of TAS as a partial replacement for cement in FC production. The work is innovative because it looks at these relationships that have been ignored before. This study intends to help create new, lightweight building materials with better qualities by focussing on the unique problems and gaps in knowledge that come up when employing TAS on FC. Understanding how TAS affects the microstructure, hydration, and rheology of FC will help us make the best use of it as a crucial part of construction materials that are good for the environment.

Novelty of the study

In this study, TAS, a by-product of lithium extraction, is explored as a novel substitute for OPC in FC. While most studies focus on standard supplemental cementitious materials, this study assesses TAS's overall effects on FC's mechanical, transport, and microstructural characteristics. The innovative approach includes detailed analyses of various TAS proportions, highlighting the optimal replacement level for maximum performance enhancement. In addition, the study explored the effects of TAS on foam concrete performance, particularly regarding foam stability, porosity reduction, and overall durability improvement. This research's repurposing of industrial waste not only advances the creation of high-performance building materials but also fits in with the growing demand for ecologically friendly and sustainable building methods.

TAS has demonstrated potential in improving the characteristics of diverse materials, such as dense concrete and ceramics. Nonetheless, its utilisation in FC has been restricted, and the methods by which it may

enhance FC qualities are predominantly unexamined. This study seeks to address this gap by investigating the interaction between TAS and the air-entrained matrix of FC, enhancing its mechanical, thermal, and durability characteristics.

TAS functions due to its elevated alumina and silica content, which, when integrated into FC, facilitates the generation of supplementary hydration products, including calcium silicate hydrate (C–S–H) gel, so improving the bonding strength and longevity of the matrix. The incorporation of TAS into the porous matrix of FC may alter the pore distribution, hence enhancing thermal insulation and soundproofing characteristics. We assert that the distinctive chemical composition of TAS enhances the stability of air-entrained structures, mitigating the collapse of air spaces during the curing process, a prevalent challenge in lightweight concrete systems. These mechanisms are distinctive to the interaction between TAS and FC's porous matrix, which markedly contrasts with solid concrete systems.

Aim and objectives

The aim was to evaluate the effectiveness of TAS as a sustainable partial OPC substitution in FC, focusing on enhancing its mechanical, transport, and microstructural properties. There were four objectives to be achieved to determine the optimal proportion of TAS that maximizes the compressive, flexural, and tensile strengths of FC, to assess the effect of TAS on the transport properties of FC, including permeability, water absorption, and sorptivity, to analyze the microstructural changes in FC with varying TAS content using Scanning Electron Microscopy (SEM) and to evaluate the overall performance and durability of FC incorporating TAS.

Experimental program

Materials

There were four main materials employed to fabricate FC which were fine river sand, OPC, clean tap water and foaming agent. Figure 1 illustrates the sieve analysis curves for two samples of fine river sand, labelled as Sample sand 1 and Sample sand 2. These curves are compared against the ASTM C33²¹ standard limits, which set the acceptable gradation range for fine aggregates used in construction. The x-axis of the graph represents the sieve sizes in millimetres, ranging from less than 0.15–4.75 mm. The sand's particle size distribution was examined, providing the sand particle's sizes. Values ranging from 0 to 100% represent the cumulative proportion of the total sample weight that goes through each sieve on the y-axis.

As shown, sample sand 1 fits predominantly within the specified upper and lower limits of the ASTM C33 ranges. This indicates a general conformity to the standard requirements for fine aggregate distribution. The smoothness and continuity of the curve indicate that the sand is well-graded and has a uniform distribution of particle sizes. Similarly, sample 2 falls within the ASTM C33 range, however, it exhibits a slightly different pattern, in that it contains a higher portion of fine than coarse particles. Although there is a variation in the particle gradation, it shows the importance of evaluating the unique gradation properties of various sand samples. The solid black line signifies the upper limit of the ASTM C33 specification for fine aggregate gradation, while the dashed black line represents the lower limit. Ensuring that the sand samples fall within these limits guarantees that the materials will contribute to the optimal performance of concrete.

When determining whether fine river sand is suitable for use in construction, particularly in concrete and mortar mixes, its physical characteristics are essential. Among these parameters are bulk density, apparent density, fineness modulus, specific gravity, moisture content, and crush index, all of which are detailed in Table 1. The river sand has features that show it is superior in a variety of construction applications. It performs best in

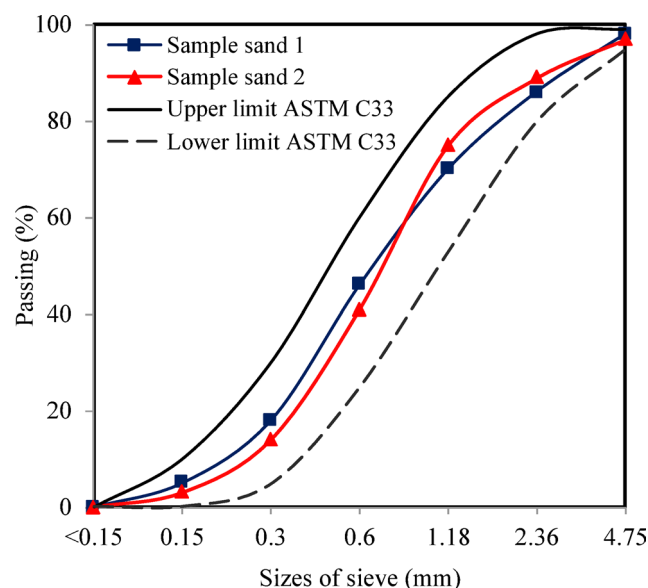


Fig. 1. Sieve analysis curve for fine river sand.

Bulk density (kg/m ³)	Apparent density (kg/m ³)	Fineness modulus	Specific gravity	Moisture content (%)	Crush index (%)
1740	2747	2.29	2.52	1.98	15.2

Table 1. Physical properties fine river sand.

Chemical constituents	OPC	TAS
Sulfur trioxide (SO ₃)	2.89	6.84
Calcium oxide (CaO)	59.91	7.52
Silicon dioxide (SiO ₂)	21.24	53.98
Aluminum oxide (Al ₂ O ₃)	5.82	22.56
Sodium oxide (Na ₂ O)	0.16	0.39
Magnesium oxide (MgO)	3.68	0.58
Iron oxide (Fe ₂ O ₃)	2.78	1.39
Potassium oxide (K ₂ O)	0.49	0.77
Loss of ignition (L.O.I)	3.03	5.97

Table 2. OPC and TAS chemical compositions.

Properties	OPC	TAS
Density (kg/m ³)	3145	3440
Specific surface area (m ² /kg)	360	340
Median particle size (pm)	30	35
Specific gravity	3.15	2.52

Table 3. OPC and TAS physical properties.

concrete and mortar mixes when its bulk density, apparent density, fineness modulus, specific gravity, moisture content, and crush index are all within ideal ranges. With these characteristics, there can be significant increase in the concrete's strength, durability, and workability, and also validate the suitability of the sand for construction.

In this study, CEM1 OPC, sourced from DRN Technologies Sdn. Bhd. in Malaysia, was used as the main binder. The chemical compositions of OPC and TAS are shown in Table 2 that exhibit significant differences, which reflect their unique properties and potential applications in construction materials. The cement has a high concentration of CaO (59.91%). The high CaO content in cement is essential for creating the calcium silicate hydrates (C–S–H) during hydration, which provides strength and stability to concrete. Cement's role is to act as a binder, which requires a substantial amount of calcium oxide to participate in the hydration reactions. However, CaO content is significantly lower in TAS (7.52%). TAS contains much lower calcium content as it is designed to interact with other compounds, such as lime or calcium hydroxide (Ca(OH)₂), to form supplementary cementitious materials (SCMs) like calcium silicate hydrate (C–S–H)^{22,23}. SiO₂ is more in TAS (53.98%) than in OPC (21.24%). TAS's high silica content can improve the pozzolanic reaction in concrete, creating more C–S–H and increasing the material's long-term strength and durability²⁴. In contrast to OPC, which has a very low percentage of Al₂O₃ (5.82%), TAS has a considerable proportion (22.56%). Concrete's early strength and resistance to sulphate attack can be improved by the calcium aluminate hydrates that can form because of TAS's high alumina concentration²². Both minerals contain trace amounts of both Na₂O and K₂O. In contrast to TAS, which has 0.39% Na₂O and 0.77% K₂O, OPC has 0.16% Na₂O and 0.49% K₂O. Concrete may expand and crack because of the alkali-aggregate reaction influenced by these alkali oxides. To prevent such reactions, TAS's somewhat greater K₂O level must be controlled²⁵. Besides, the MgO content in OPC is 3.68%, whereas in TAS it is 0.58%. The levels of MgO in both materials are quite small and are not likely to have a major impact, even if they can help produce expansive compounds like brucite in concrete²². The Fe₂O₃ level in TAS is 1.39%, whereas it is 2.78% in OPC. The lower iron content in TAS can influence the colour and may provide additional pozzolanic activity, contributing to the strength and durability of concrete²². L.O.I represents the amount of weight loss when the material is heated to a high temperature, indicating the presence of volatile compounds. TAS has a marginally greater L.O.I. of 5.97% than OPC, which has an L.O.I. of 3.03%. The breakdown of leftover organic compounds and carbonates during the heating process may be the cause of this increased L.O.I. in TAS²³. The unique characteristics and possible applications in building of OPC and TAS are highlighted by their chemical compositions. TAS's high SiO₂ and Al₂O₃ levels imply that it may be used as a supplemental ingredient to enhance the strength and durability of concrete, whereas OPC's high CaO and SO₃ values make it perfect as a primary binder.

Table 3 displays a comparison of the physical characteristics of OPC and TAS, showing the variation in their properties, which may impact their effectiveness in concrete mixtures. One notable property is density. OPC has a density of 3145 kg/m³, while TAS has a higher density of 3440 kg/m³. The concrete mix's strength and

durability may be improved by this higher TAS density, which could also raise the mix’s total density²⁶. For OPC and TAS, the specific surface area, a measure of the total surface area accessible for reaction per unit mass, is 360 m²/kg and 340 m²/kg, respectively. A higher specific surface area generally indicates better reactivity with water and other components in the mix, influencing the rate of hydration and strength development. Despite TAS having a slightly lower specific surface area than OPC, its higher silica content may still contribute to significant pozzolanic activity²⁷. The OPC has a finer median particle size of 30 µm compared to 35 µm for TAS. Finer particles usually lead to a denser and more homogeneous mix, which can enhance the mechanical properties of the concrete. However, TAS’s marginally bigger median particle size might be beneficial since it improves the microstructure of the concrete and offers a superior filler effect²⁸. OPC’s specific gravity is 3.15, whereas TAS’s is 2.52. TAS’s lower specific gravity suggests that it is less dense than OPC on a volumetric level, which may have an impact on the mix ratios and total weight of the concrete. In structural applications, this feature is crucial when weighing the entire material. It can also affect the mix design to have the right workability and strength²⁹. The physical characteristics of OPC and TAS differ somewhat, but each offers advantages that can be used to improve the total concrete performance. In certain situations, TAS’s lower specific gravity can have an impact on mix design by providing a lighter substitute for OPC³⁰. TAS exhibits notable pozzolanic activity primarily due to its high silica (SiO₂) and alumina (Al₂O₃) content, which contribute to the formation of calcium silicate hydrate (C–S–H) gel during the hydration process. Iron oxide (Fe₂O₃), although present in higher amounts in TAS compared to other materials, is considered inert in terms of pozzolanic reactivity and does not directly influence the pozzolanic effects of the material.

Foaming agents generally influence the properties of FC, particularly with the stability and density of the foam, and overall strength of the concrete. Table 4 presents the characteristics of EABASSOC, a particular protein-based foaming agent that is frequently employed in the FC production. The protein-based group includes the foaming agent EABASSOC. Comparing protein-based foaming agents to synthetic foaming agents, the former are known to have greater mechanical qualities and durability due to their smaller pore sizes and higher foam stability³¹. It can produce foam that expands to 20 times its initial liquid volume since the EABASSOC foaming agent has an expansion ratio of 20x. Producing FC with a constant density and superior thermal insulation qualities is made possible by this high expansion ratio¹⁹. Also, the pH of the foaming agent is 5.64 ± 0.1, which indicates a mild acidity. In order to ensure steady foam production and stability throughout the mixing process, this pH level aids in maintaining the integrity of the protein structure in the foaming agent³². The foaming agent has a density of 65 ± 5 kg/m³. In order to improve the concrete mix’s thermal and acoustic insulation qualities, low-density foaming agents are recommended for lightweight concrete production³³. The density of the foaming agent in relation to water is indicated by its specific gravity, which is 1.11. A specific gravity slightly higher than water ensures that the foaming agent can mix well with water and other concrete components, creating a uniform foam distribution in the mix³⁴. The foaming agent is pale brown in colour; a typical characteristic of protein-based foaming agents derived from natural sources such as animal proteins or plant residues. While the colour does not significantly affect the performance of the foaming agent, it may indicate its natural origin. The properties of the EABASSOC protein-based foaming agent, such as its high expansion ratio, suitable pH, low density, and appropriate specific gravity, contribute to its effectiveness in producing high-quality FC. This agent produces foam that is stable and consistent, which improves the final concrete product’s mechanical qualities and durability, making it a popular option for a variety of building applications. The foam expansion ratio was verified using the graduated cylinder method, where the foam volume before and after expansion was measured to determine the expansion ratio. The expansion ratio is calculated as the volume increase of the foam after incorporation into the mortar. The stable foam was merged into the base mix using a mechanical mixer set at a mixing speed of 50 RPM. The foam was added gradually to the mortar to ensure uniform distribution and optimal integration. The flowability of the base mortar is a critical parameter that can affect the final quality of FC. We have now included a description of the flowability range of the base mortar before the foam is added. Before the addition of foam, the base mortar demonstrated a flowability range of 220–250 mm, as assessed by the slump test. This flowability range guarantees the mortar’s workability and facilitates adequate foam integration.

The chemical compositions of TAS and OPC are depicted in a ternary diagram in Fig. 2, and their physical characteristics are shown in Fig. 3. The primary chemical components, SiO₂, CaO, and Al₂O₃, are represented by the triangle’s three corners. With their proportions decreasing along the axes toward the opposing sides, each corner represents a 100% concentration of the corresponding component. This diagram shows TAS as a green dot and OPC as a red dot. The green dot, representing TAS, is located closer to the SiO₂ corner, indicating a high Silicon Dioxide content of approximately 53.98%. It is also nearer to the Al₂O₃ corner than to the CaO corner, showing significant amounts of Al₂O₃ (22.56%) and a lower Calcium Oxide concentration (7.52%). This

Properties	Descriptions/values
Brand	EABASSOC
Group	Protein-based
Expansion	20×
pH value	5.64 ± 0.1
Density (kg/m ³)	65 ± 5
Specific gravity	1.11
Color	Pale brown

Table 4. Foaming agent properties.

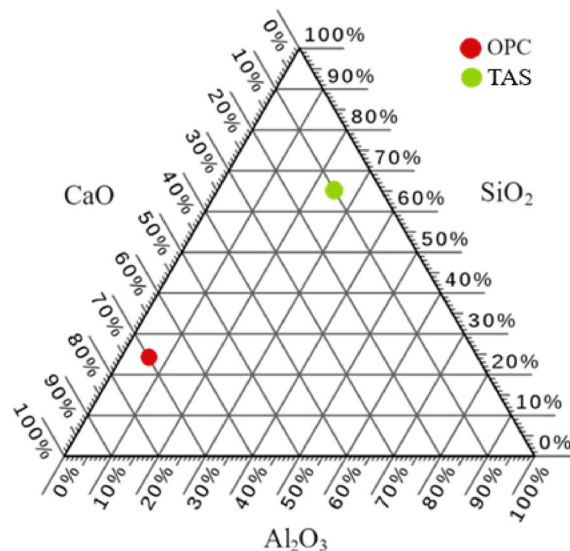


Fig. 2. A ternary diagram illustrates the chemical constituents of TAS and OPC.



Fig. 3. Physical appearance of TAS (left) and OPC (right).

positioning implies that TAS is rich in silica and alumina, which makes it suitable for applications requiring high thermal and chemical stability. On the other hand, A high CaO content (34.7%) is indicated by the red dot, which stands for OPC, being located closer to the CaO corner. Its location farther away from the SiO₂ and Al₂O₃ corners suggests that there is less silicon dioxide (2.61%) and al₂O₃ (0.26%) present. OPC exhibits this composition, which uses calcium compounds to help concrete hydrate and build mechanical strength. The distinct placements of TAS and OPC in the ternary diagram highlight their different chemical compositions and properties. TAS, with high SiO₂ and Al₂O₃ contents, could be beneficial as a supplementary cementitious material, enhancing durability and strength when mixed with traditional cement. Additionally, high CaO content of OPC is vital for its function as the main binder in concrete, which is necessary for structural applications and strength development. Because TAS has a high silica and alumina concentration, it can be used in ceramics, refractory materials, and as an additive in concrete to increase durability and resistance to chemical attacks. OPC is the fundamental ingredient in the formation of concrete and is still essential in construction because of its ideal composition for hydration and strength development. The different chemical compositions of TAS and OPC are clearly depicted in this ternary diagram, which also provides information about their individual characteristics and possible uses in the construction sector.

Mix design

The mix design for FC targeting a density of 1000 kg/m³ includes diverse proportions of TAS as a substitution for OPC. The mix uses a cement-to-sand ratio of 1:1.5 and a water-to-cement ratio of 0.46 to ensure proper workability and strength. The TAS was substituted for cement at levels: 0%, 10%, 20%, 30%, and 40%. Table 5 presents the details of the FC mix design, where each mix indication (FC0, FC10, FC20, FC30, and FC40) denotes a particular percentage of TAS substitution. For instance, FC0, the control mix with no TAS, contains 373.4 kg/m³ of OPC and 560.1 kg/m³ of sand. As the TAS substitution increases, the amount of OPC decreases proportionately while keeping the total binder content constant. For example, FC10 includes 37.34 kg/m³

Mix	Density	TAS	TAS	OPC	Sand	Water	Foam
Designation	(kg/m ³)	(%)	(kg/m ³)	(kg/m ³)	(kg/m ³)	(kg/m ³)	(kg/m ³)
FC0	1000	0	0	373.4	560.1	171.8	30.66
FC10	1000	10	37.34	336.1	560.1	171.8	30.66
FC20	1000	20	74.68	298.7	560.1	171.8	30.66
FC30	1000	30	112.02	261.4	560.1	171.8	30.66
FC40	1000	40	149.36	224.1	560.1	171.8	30.66

Table 5. Mix design of FC containing different weight fractions of TAS.

of TAS and 336.1 kg/m³ of OPC, while FC40 has 149.4 kg/m³ of TAS and 224.1 kg/m³ of OPC. In order to measure the effect of TAS on the concrete properties, the sand, water, and foam quantities were kept constants for all mixes. Each mix maintains a consistent density of 1000 kg/m³ and a uniform sand content of 560.1 kg/m³. The water content is fixed at 171.8 kg/m³, ensuring a constant water-to-cement ratio across all mixes. The foam content is also kept steady at 30.66 kg/m³ to achieve the desired lightweight properties of the FC. This design makes it possible to compare the durability and mechanical characteristics of FC with variable cement replacement amounts by altering the TAS content. With the goal of improving the material’s performance and sustainability, this mix design technique methodically assesses the effects of TAS as a partial cement substitute in FC. Any observed differences in the FC’s properties are guaranteed to be caused by the different degrees of TAS replacement because sand, water, and foam are used consistently in all mixes.

FC preparation

A target density of 1000 kg/m³ was considered for the FC preparation, which entailed a careful process ensuring quality and consistency of the mix. This target density is achieved by carefully controlling the amount of stable foam introduced into the mortar slurry. A mix design ratio of 1:1.5 for cement and sand, and water/cement ratio of 0.46 was adopted. The influence of varying TAS amounts as an OPC substitute is examined in this study. A dry mixture of OPC, sand, and TAS is used to start the mixing process. Three minutes are spent completely mixing and combining the dry ingredients. This stage guarantees that the TAS, sand, and cement are dispersed equally throughout the mixture, which is essential for obtaining a homogenous blend. After combining the dry ingredients completely, water is added to the mixture. The blending of the mix continued for another 4 min, ensuring a uniform distribution of water and adequate hydration of all constituents. The workability of the mix was tested using a flow table. The test is used to determine the mix’s homogeneity and fluidity and helps to ensure mix is smoothly poured into the formwork. Stable foam is created using the Portafoam TM-2 machine, which employs the EABASSOC protein-based foaming agent. This protein foaming agent is known for producing stable and uniform bubbles, which contribute to the lightweight nature and insulating properties of the FC. The required quantity of foam is then added to the prepared mortar slurry. Precise control is maintained over the quantity of foam supplied in order to attain the desired density of 1000 kg/m³. This phase is vital as it directly influences the density and strength of the FC. After adding foam, a further five minutes of blending was done to ensure a perfect mix. This extended mixing period ensures that the foam is uniformly distributed throughout the mortar slurry and that there are no visible foam pockets. The goal is to achieve a consistent and homogeneous mix with stable air bubbles evenly dispersed. After that, the FC samples are left to cure in dampness till the test day. For the cement to completely hydrate and the concrete’s mechanical properties to fully develop, moisture curing is essential. This procedure aids in preserving the FC’s intended strength and longevity. In conclusion, a number of exact procedures are involved in the preparation of FC to guarantee the consistency and caliber of the finished product. The goal of the study is to create FC with the best density, workability, and mechanical qualities while examining the effects of TAS as a cement substitute by carefully regulating the ingredient amounts and the mixing procedure.

Experimental setup
Fresh-state properties

Assessing the workability, setting durations, and fresh density of FC is crucial for comprehending its performance during placement and its capacity to fulfil the demands of diverse building applications. These qualities affect the manageability, efficiency of the construction process, and the ultimate performance of the concrete. A variety of tests are utilised to evaluate these qualities, each offering significant insights into the properties of fresh concrete and their influence on later phases of the construction process.

The slump flow table test, performed in accordance with ASTM C230-97³⁵, is frequently utilised to evaluate the workability and consistency of fresh FC. This test assesses the ease with which concrete may disperse on a level surface, indicating its fluidity and installation convenience. The process entails positioning a sample of fresh concrete into a conical form on a flow table. Upon the removal of the cone, the concrete is let to disperse, and the diameter of the dispersion is recorded. A bigger spread indicates enhanced workability of the concrete. This test is essential for assessing the concrete’s capacity to flow and occupy moulds or formwork without requiring excessive compaction or vibration, hence potentially lowering labour costs and enhancing efficiency in the construction process. The slump flow test indicates the concrete’s capacity for placement and compaction under its own weight, particularly in scenarios necessitating considerable fluidity, such as thin sections or intricate formwork. The test also offers insights into the likelihood of segregation or bleeding, which may jeopardise the

final quality of the concrete. Engineers can evaluate the slump flow to ascertain the mix's suitability for certain applications, guaranteeing optimal performance of the concrete during installation and subsequent curing.

The initial and final setting times of fresh FC are ascertained by the setting time test, performed in accordance with BS EN 196-3³⁶. The setting time is a crucial factor in concrete workability, as it determines the period during which fresh concrete stays malleable and can be shaped or positioned before it commences hardening. The first setting time is the moment when concrete shifts from a liquid to a semi-solid state, whereas the final setting time indicates when the concrete has completely hardened and is no longer malleable. Comprehending these durations is crucial for construction scheduling, as it aids in assessing the time allotted for placement, finishing, and modifications prior to the concrete becoming unmanageable. The setting times significantly influence the scheduling and coordination of building activities. If the setting durations are excessively brief, the concrete may cure prematurely, complicating the placement or finishing of the material. Conversely, protracted setup times may impede construction efforts, necessitating prolonged labour hours. Engineers can optimise the mix design by evaluating both the start and ultimate setting times to attain an ideal equilibrium between workability and durability, while also assuring optimal use of labour and resources.

The fresh density of FC is assessed in line with BS 12,350-6³⁷, offering insight into the material's mass per unit volume. This test entails measuring a specified volume of fresh concrete to ascertain its density, a vital indicator of the material's uniformity and efficacy. The fresh density significantly influences the ultimate characteristics of hardened FC, including its strength and durability. It also influences the general workability of the concrete mix and affects the convenience of transporting, placing, and finishing the material during construction. In FC, the fresh density is crucial since it affects the material's thermal and acoustic insulation qualities, as well as its mechanical performance. A reduced fresh density generally signifies an increased amount of entrained air, perhaps enhancing insulating capabilities while diminishing strength. Conversely, an increased fresh density signifies a more compact concrete with more strength potential, although may diminish insulating characteristics. Consequently, it is essential to ascertain the fresh density to guarantee that the mixture satisfies the precise criteria for the planned application, balancing strength, workability, and other requisite attributes like as thermal and acoustic insulation.

Transport properties

To achieve a thorough comprehension of the transport qualities of concrete, multiple experimental techniques are utilised to assess different facets of its performance throughout time. These methods are meticulously selected to evaluate the material's response to water absorption, air permeability, porosity, and chloride ion penetration, which are critical indications of durability and long-term stability in varying environmental circumstances.

Water absorption tests are conducted at 7, 14, and 28 days, following the protocols specified in BS 1881-122³⁸. Cylindrical specimens having a diameter of 50 mm and a height of 100 mm are manufactured for this test. The process entails submerging the specimens in water and quantifying the overall water absorption. This test is essential for assessing the material's moisture absorption capacity, which can considerably influence its long-term performance. Water absorption is crucial for concrete's resilience to environmental conditions, including freeze-thaw cycles and chemical assaults, and serves as a reliable measure of the material's overall longevity. The intervals of 7, 14, and 28 days offer insights into the progression of water absorption over varied curing durations, which is crucial for assessing the long-term performance of concrete subjected to diverse moisture circumstances.

The intrinsic air permeability of the concrete is evaluated according to the methodology established by Cabrera and Lynsdale³⁹, utilising FC cylinders of 45 mm in diameter and 50 mm in height. This test assesses the permeability of air through concrete, which correlates directly with its pore structure. The air permeability is a critical measure of a material's durability, as it indicates the ease with which moisture and gases can infiltrate the concrete, potentially resulting in reinforcement corrosion or other degrading forms. The air permeability test yields critical information regarding the connectivity and distribution of pores in concrete, influencing its mechanical qualities and resilience against harsh environmental factors.

Porosity measurements are performed at 7, 14, and 28 days utilising cylindrical specimens (50 mm in diameter and 50 mm in height), in accordance with the RILEM TC-14 CPC11⁴⁰ standards. Porosity is an essential characteristic of concrete that profoundly influences its strength, durability, and permeability. This approach quantifies the total void area in the concrete, which is essential for assessing its resistance to the infiltration of deleterious elements such as water and chloride ions. Elevated porosity is frequently linked to diminished strength and heightened susceptibility to environmental degradation, especially in coastal or industrial settings. Monitoring porosity at various curing ages allows us to evaluate the temporal evolution of the pore structure and its influence on the concrete's overall durability.

The Rapid Chloride Permeability Test (RCPT) is conducted in accordance with ASTM C1202⁴¹ to assess the concrete's resistance to chloride ion infiltration. This test used a FC disc with a diameter of 100 mm and a thickness of 50 mm. During the examination, an electric charge is transmitted through the specimen, and the quantity of charge that traverses is quantified. This indicates the permeability of the concrete to chloride ions. The penetration of chloride ions is a crucial determinant in evaluating the durability of concrete, especially in settings subjected to de-icing salts or maritime environments. The RCPT is a recognised technique for assessing concrete's resistance to chloride-induced corrosion, a principal factor in the degradation of reinforcement within concrete structures. The test predicts the concrete's performance in hostile settings and its possible service life by assessing permeability.

To guarantee the consistency and reliability of the test results, all transport property tests are performed on an average of three specimens, adhering to established standards for sample preparation and testing. This method reduces specimen variability and guarantees that the data accurately reflects the material's overall qualities.

The utilisation of several specimens facilitates a more comprehensive investigation of the concrete's transport qualities, enhancing the reliability of the results' interpretation.

Mechanical properties

Assessing compressive, flexural, and tensile strengths, in addition to the modulus of elasticity, is essential for ascertaining the comprehensive mechanical qualities and structural capability of concrete. These qualities are crucial for guaranteeing that concrete functions effectively in many applications and endures the varied loads it may face during use.

The compressive strength of concrete is assessed using $100 \times 100 \times 100$ mm cube specimens, following BS 12,390-3⁴². This test assesses the maximum load that concrete can withstand prior to failure, serving as a crucial indicator of its capacity to support axial loads. Compressive strength is a prominent test for concrete, as it directly correlates with the material's structural efficacy. Elevated compressive strength is essential for concrete's capacity to endure substantial loads without distortion or fracture. The test is performed at several curing ages, specifically 7, 14, and 28 days, to evaluate the progression of strength over time, facilitating an analysis of the concrete's maturation and hardening under diverse environmental conditions. The outcomes of this test are essential for structural design, as they assist engineers in ascertaining the suitable mix and grade of concrete for construction applications.

Flexural strength is evaluated with prism specimens measuring $100 \times 100 \times 500$ mm, in line with BS 12,390-5⁴³. This test evaluates concrete's capacity to withstand bending strains, which is crucial for applications involving flexural loading, including beams, slabs, and pavements. The test entails exerting a load at the midpoint of the prism until failure, with the flexural strength determined by the maximum load and the specimen's dimensions. This feature is essential for enabling concrete buildings to endure bending forces without fracturing, particularly in regions with unequal load distribution. The flexural strength test indicates the concrete's performance under actual conditions, when bending and flexural forces frequently occur.

The splitting tensile strength is assessed using cylindrical specimens measuring 100×200 mm in accordance with BS 12,390-6⁴⁴. This test assesses the tensile strength of concrete, crucial for comprehending its performance under tensile loading circumstances. Tensile strength, as contrast to compressive strength, evaluates the material's resistance to pressures that endeavour to elongate or separate it. This feature is crucial for concrete constructions experiencing tensile strains, such as bridge decks or pavements. The splitting tensile test entails exerting a compressive force along the cylinder's diameter, resulting in longitudinal splitting. The outcome offers an assessment of the concrete's capacity to withstand cracking from tensile stresses, which is essential for the structure's durability and longevity.

The modulus of elasticity is evaluated using cylindrical specimens of 100×200 mm, in compliance with ASTM C469⁴⁵. This attribute quantifies the stiffness of the concrete, reflecting its deformation under an applied load. The modulus of elasticity is a crucial quantity in structural design, since it indicates the concrete's capacity to withstand deformation under stress. Concrete possessing a greater modulus of elasticity will exhibit increased stiffness, resulting in less deformation under load, which is crucial for structures necessitating high dimensional stability, such as skyscrapers or bridges. The test is applying a specified compressive load to the cylinder and measuring the resultant deformation, from which the modulus of elasticity is derived. This feature is essential for assessing the concrete's performance under service loads, especially when it is integrated into a structural system comprising other materials like steel or composite elements.

Microstructural characteristics

Microstructural study is essential for comprehending the behaviour and performance of concrete at the microscopic scale. Analysing the material's microstructure provides critical insights into how the composition and structural arrangement of its constituents influence the overall characteristics, durability, and long-term performance of the concrete. This study utilises scanning electron microscopy (SEM) to meticulously analyse the microstructure of concrete, providing a high-resolution perspective on the material's pore architecture, constituent distribution, and any flaws. Pore size distribution measurements are conducted on FC to further examine the influence of pore features on the material's transport capabilities and durability.

Microstructural study is performed via a Quanta FEG 650 microscope and an Oxford Instruments X-Max microscope, both featuring sophisticated imaging capabilities to capture intricate features of the concrete's structure. Samples measuring $15 \text{ mm} \times 15 \text{ mm} \times 15 \text{ mm}$ are meticulously prepared for SEM examination. The sample dimensions are selected to enable adequate representation of the area of interest, facilitating a thorough evaluation of the microstructure. The specimens are analysed at a magnification of 120x, allowing for adequate detail to discern critical elements such as pore structure, ingredient distribution, and any defects or fissures that may affect the concrete's performance.

The SEM technique provides an effective means of obtaining high-resolution images of the interior structure of FC. At this magnification, the SEM can disclose vital information on pore size and distribution, which are crucial determinants of the material's permeability, strength, and durability. Moreover, SEM offers insights into the distribution and bonding of several components, including aggregates, binders, and additives, which significantly influence the concrete's overall performance under varying environmental circumstances. Analysing the microstructure allows us to detect potential defects, like vacancies, cracks, or inadequately bound particles, which could result in diminished performance or premature breakdown of the concrete under stress or in harsh settings.

The pore structure is a crucial element of the microstructure revealed using SEM. The pores in concrete determine its permeability, and their size, distribution, and connection directly affect the material's resistance to water absorption, chloride intrusion, and chemical degradation. SEM examination facilitates the detection of several pore types, including capillary pores, air voids, and gel pores, which can yield insights into the hydration

process and the efficacy of the binding phases within the concrete. A well-developed microstructure characterised by a more refined and less linked pore network generally signifies enhanced durability and strength.

The distribution of the elements is another essential characteristic noticed using SEM. The appropriate distribution of aggregates, cementitious substances, and additives enhances the consistency and durability of the concrete. SEM facilitates the detection of discrepancies in this distribution, such as irregular adhesion between the binder and aggregate particles, which may compromise the concrete's overall efficacy. Furthermore, the investigation can elucidate the interactions of different additives or extra materials, such as fly ash, slag, or silica fume, with the cementitious matrix, yielding critical insights into their contributions to the enhancement of concrete qualities.

Pore size distribution studies of FC are performed using a mercury intrusion porosimetry (MIP) equipment to examine the pore structure of the concrete in greater detail. This technique is extensively employed to evaluate the pore architecture of porous substances. The distribution of pore sizes is crucial in assessing the permeability, strength, and durability of FC, as the dimensions, connectedness, and arrangement of pores can profoundly influence its transport characteristics.

The instrument utilised for this study is the Autopore IV 9500 (Micromeritics), engineered for high-resolution assessment of pore dimensions from 3 nm to 360 μm . During this procedure, mercury is introduced into the concrete specimen under progressively increasing pressure, and the volume of mercury that permeates at each pressure increment is recorded. This data is utilised to compute the distribution of pore diameters within the specimen. The findings furnish comprehensive data regarding total pore volume, mean pore diameter, and pore size distribution. The pore size distribution elucidates the extent of connectivity among pores and the capacity for fluid transport inside the material, which directly influences the concrete's resistance to water, chloride ions, and other deleterious substances. A precisely regulated pore size distribution is frequently linked to enhanced performance regarding strength, permeability, and durability.

Besides elucidating the advantageous features of the concrete's microstructure, SEM is also proficient in detecting probable flaws or deficiencies. These may encompass the existence of fissures, voids, or regions of inadequate adhesion that could undermine the material's strength and longevity. Improper hydration, insufficient curing, or incorrect mixing can result in the development of weak zones or air pockets, so diminishing the overall integrity of the concrete. Early identification of these defects can guide decisions on the optimisation of mix designs, curing techniques, and materials to enhance the overall performance of the concrete.

The SEM analysis is performed in compliance with ISO 16,700⁴⁶ standards, guaranteeing that the techniques and outcomes align with globally acknowledged principles for microstructural characterisation. This standard establishes a framework for the preparation of concrete samples and the corresponding procedures for imaging and analysis, guaranteeing reliable and reproducible findings. By adhering to these guidelines, we can guarantee that the SEM analysis yields a rigorous and scientifically valid evaluation of the concrete's microstructure, hence enhancing the overall quality and dependability of the research outcomes.

Results and discussion

Fresh-state properties

Slump flow

The impact of using TAS as a substitute for cement on the slump diameter of various FC mixtures is depicted in Fig. 4. The slump flow test, performed following ASTM C230-97 standards, assesses the workability and fluidity

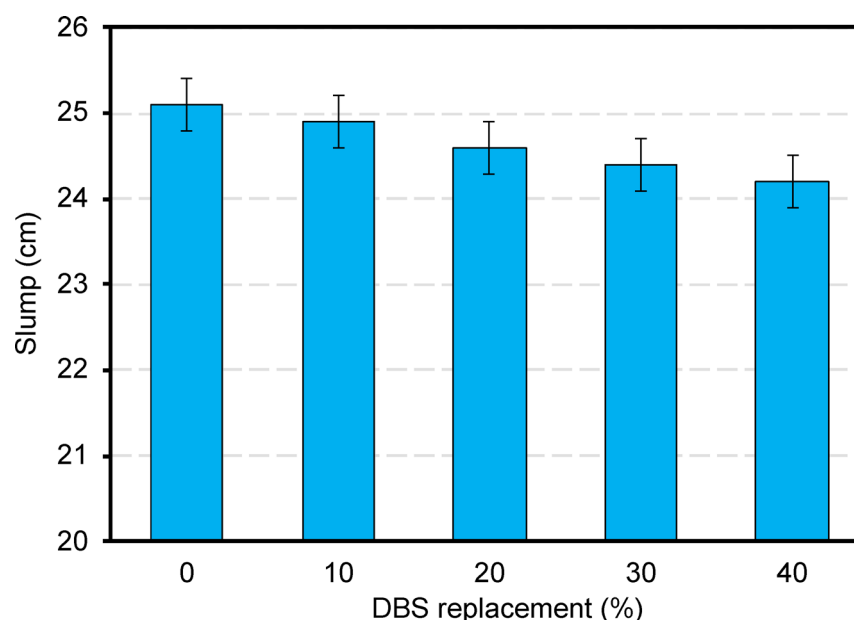


Fig. 4. Influence of TAS cement replacement on slump diameter of different mixes.

of the fresh concrete mixture. The findings reveal a noticeable shift in slump diameter as the TAS replacement percentage rises. For the control mix (FC0) with 0% TAS replacement, the slump diameter is relatively high, indicating excellent workability. As the TAS content increases to 10% (FC10), the slump diameter slightly decreases, hinting at a minor reduction in workability. This trend continues as the TAS replacement levels increase to 20% (FC20), 30% (FC30), and 40% (FC40), resulting in further decreases in slump diameters. This reduction can be attributed to the differences in the physical characteristics and chemical make-up of TAS compared to OPC. The higher silica content and lower calcium content in TAS may affect the water demand and the rheology of the mix, leading to decreased flowability²⁴. These results align with other studies that explored the use of alternative cementitious materials. For instance, research on using masonry residue and ground copper slag as partial cement replacements in self-compacting concrete has demonstrated similar trends in reduced slump flow with increased replacement levels^{47,48}. In summary, incorporating TAS as a partial OPC replacement in FC affects workability, but the changes in slump flow remain within acceptable ranges for practical use. Adjustments to mix proportions or the use of superplasticizers might be necessary to achieve desired workability levels in mixes with higher TAS content.

Hardened density

Figure 5 presents the effect of TAS as a replacement for cement on the hardened density of various FC mixes. The y-axis denotes the hardened density in kg/m^3 , while the x-axis displays the percentage of TAS replacement, from 0 to 40%. The graph's bars represent the average hardened density for each mix, accompanied by error bars indicating measurement variability. Unexpectedly, the hardened density of the FC increases with higher TAS replacement levels. The control mix, with 0% TAS replacement, has a hardened density of about 1000 kg/m^3 . As the TAS content rises to 10%, 20%, 30%, and 40%, the hardened density increases, reaching approximately 1025 kg/m^3 at 40% TAS replacement. This pattern indicates that incorporating TAS, rather than reducing the density as might be expected with lighter additives, enhances the FC's density. This surprising outcome may be attributed to the unique properties of TAS and its interaction with the cement matrix. One possible reason is that TAS improves the packing density of the mix or contributes to additional hydration products, thereby increasing the concrete's overall mass and density. This finding contrasts with other studies on alternative materials like ground waste clay brick and rice husk ash, which generally reduce concrete density due to their lower specific gravities compared to OPC⁴⁹. In conclusion, Fig. 5 illustrates that increasing TAS content in FC leads to a rise in its hardened density. This intriguing result suggests that TAS uniquely influences the material properties of FC, meriting further investigation to fully understand the mechanisms involved and optimize the mix design for specific density and mechanical property requirements.

Setting time

Figure 6 illustrates how the setting time of various FC mixes is affected by replacing cement with TAS. The y-axis indicates the setting time in minutes, while the x-axis shows the percentage of TAS replacement, ranging from 0 to 40%. The initial and final setting times are shown for each mix. For the control mix (0% TAS replacement), the initial setting time is about 100 min, and the final setting time is approximately 175 min. As the TAS replacement level increases, both the initial and final setting times exhibit a noticeable trend. With a 10% TAS replacement, the initial setting time decreases slightly, while the final setting time increases. At 20% TAS replacement, the initial setting time remains relatively unchanged, but the final setting time continues to rise. This trend becomes more evident at 30% and 40% TAS replacement, with significant increases in both initial and final setting times.

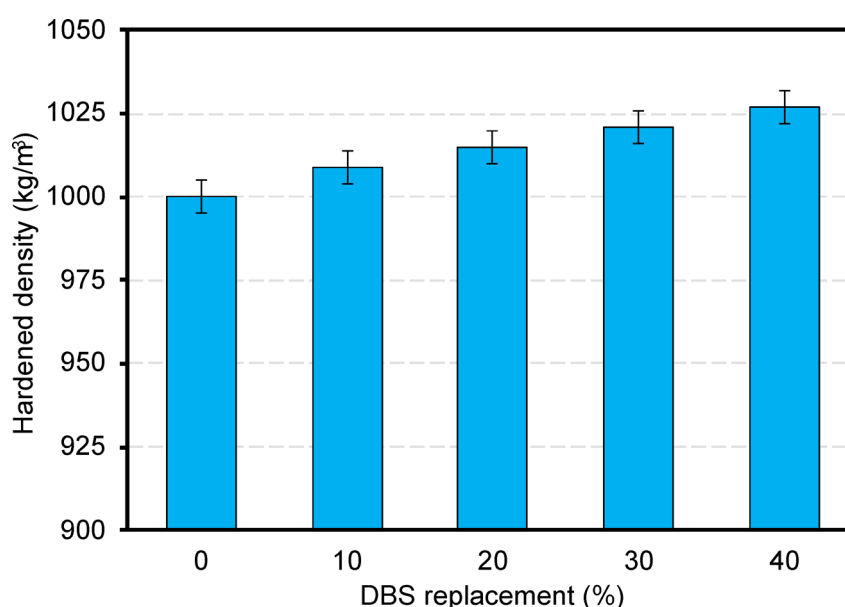


Fig. 5. Influence of TAS cement replacement on hardened density of different mixes.

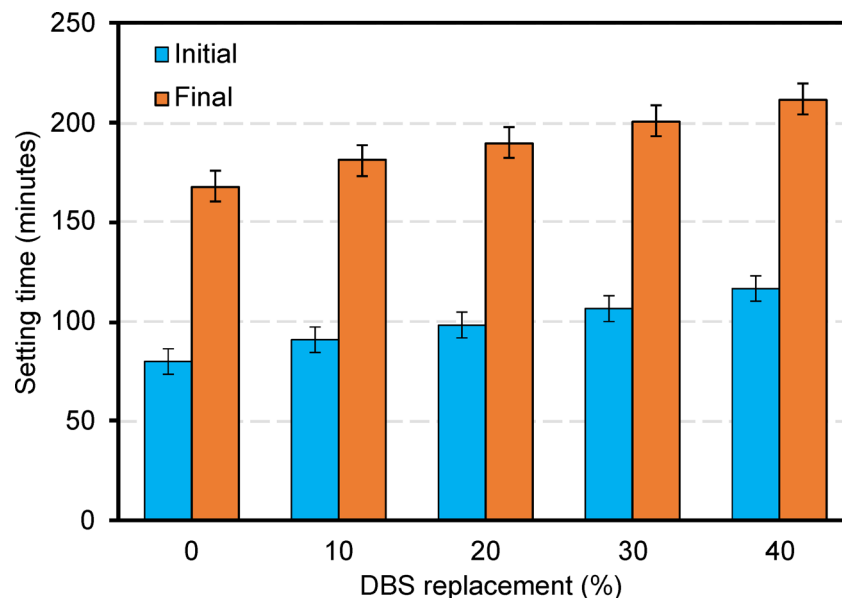


Fig. 6. Influence of TAS cement replacement on setting time of different mixes.

At 40% TAS replacement, the final setting time extends up to 200 min. The increased setting time with higher TAS content is due to the lower reactivity of TAS compared to OPC. As a by-product of lithium extraction, TAS contains high levels of silica and alumina, which react more slowly with water than the calcium silicates in OPC. This slower reaction rate delays hydration, leading to prolonged setting times²⁴. These results are consistent with other studies exploring alternative cementitious materials. For instance, using fly ash and slag as partial replacements for OPC has also been shown to extend concrete's setting time due to their slower hydration rates. In conclusion, Fig. 6 shows that increasing the TAS content in FC leads to longer setting times. This extended setting period can be beneficial for applications that require longer workability, allowing more time for mixing, transporting, and placing the concrete. However, it is essential to balance the extended setting times with the overall project schedule and requirements to ensure efficient construction processes.

The study noted that augmenting the TAS content in FC results in an elongation of the setting time. This delay may improve workability and provide more time for placement, but it also presents significant disadvantages in specific building scenarios. In bulk casting, extended setting durations might result in delayed hardening and heighten the danger of differential curing, which may compromise the final strength and durability of substantial pours. In warmer weather circumstances, extended setting durations might increase the danger of drying shrinkage or uneven curing, as the mixture may remain in a plastic state for longer, subjecting it to external influences. Conversely, in cold conditions, extended setting durations might be beneficial as they facilitate improved hydration and adhesion among the components, ensuring more thorough curing at reduced temperatures. Nonetheless, this may also postpone the construction timeline, resulting in prolonged curing durations and possible interruptions in the production.

To alleviate potential delays resulting from extended setting time, the incorporation of specific chemical activators, including calcium-based accelerators or alkali activators, can expedite the setting time of FC while maintaining the mix's overall strength and durability. These activators can improve the hydration process, particularly in bulk casting at elevated temperatures where swift setting is essential. Additionally, by decreasing the water-to-binder ratio may facilitate the regulation of the setting time by diminishing the quantity of water accessible for hydration, hence abbreviating the curing duration. Nonetheless, this must be meticulously balanced to prevent undermining the workability and consistency of the mixture.

Transport properties

Porosity

Figure 7 illustrates the effect of replacing cement with TAS on the permeable porosity of FC over different curing periods (7, 28, 56, and 90 days). The data show a consistent trend of decreasing porosity as the percentage of TAS replacement increases. For example, at 40% TAS replacement, the porosity decreases from approximately 50% on Day 7 to around 38% by Day 90. This suggests that TAS effectively fills the voids within the concrete matrix, resulting in a denser structure and reduced overall pore volume. Furthermore, extended curing times contribute to reductions in porosity across all levels of TAS replacement. This is due to the ongoing hydration process that forms more C–S–H gel, which fills the pores and improves the concrete's microstructure. The reduction in porosity enhances the durability and mechanical properties of FC, highlighting TAS as a promising material for sustainable construction. Supporting studies, such as those by Nair & Viswanathan⁴⁹, have also found that using alternative materials to replace cement can improve concrete's durability and decrease its permeability.

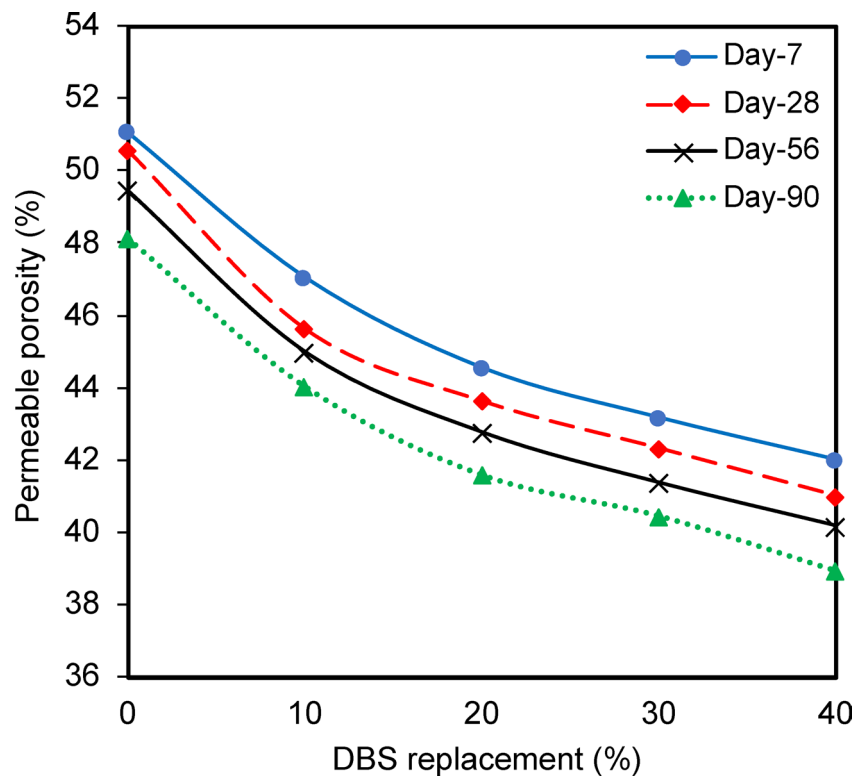


Fig. 7. Influence of TAS cement replacement on permeable porosity of different mixes.

Water absorption

The water absorption of concrete is a vital characteristic that affects its durability and performance, particularly in severe weather circumstances. The water absorption of concrete is influenced by the mix design, curing circumstances, and the type of aggregate utilised. Recycled aggregates, obtained from the recycling of discarded concrete, can have more porosity than natural aggregates. The augmented porosity results in elevated water absorption rates, as the pores in the recycled aggregates often retain more water than the denser, less porous natural aggregates⁵⁰.

The impact of TAS replacement on the water absorption of FC over different curing periods (7, 28, 56, and 90 days) is shown in Fig. 8. The data reveal that as the percentage of TAS replacement increases, water absorption consistently decreases across all curing times. For instance, at 40% TAS replacement, water absorption decreases from approximately 23% on Day 7 to around 18% by Day 90. This trend indicates that TAS effectively enhances the FC microstructure and enhances its durability by reducing water absorption. Longer curing times contribute to further reductions in water absorption due to the continued formation of C–S–H gel, which fills the pores and improves the concrete's impermeability. Supporting studies by Nair & Viswanathan⁴⁹ have similarly found that using alternative materials to replace cement can significantly reduce water absorption and porosity, thereby enhancing the mechanical properties and durability of the concrete. These findings highlight the potential of TAS as a sustainable cement replacement material that improves the performance and longevity of FC.

Intrinsic air permeability

Figure 9 depicts the effect of substituting TAS for cement on the intrinsic air permeability of FC. As the proportion of TAS increases from 0 to 40%, there is a significant reduction in air permeability. For the control mixture (0% TAS), the air permeability is around $0.4 \times 10^{-9} \text{ m}^2/\text{s}$, which decreases to approximately $0.25 \times 10^{-9} \text{ m}^2/\text{s}$ at a 40% TAS replacement. This reduction signifies that a higher TAS content effectively diminishes the pore connectivity within the concrete, thereby increasing its density and restricting air movement through the material. Supporting research, including that by Othman et al.⁵¹ underscores the advantages of utilizing industrial wastes like TAS in cement replacement. Their study indicates that TAS significantly enhances the impermeability and durability of FC by reducing pore connectivity. Similarly, Pizoni⁵² found that using recycled foam concrete dust as a partial cement substitute also lowered permeability in cement composites, which aligns with the observed improvements in air permeability with increased TAS content. These studies illustrate that TAS effectively enhances the durability and performance of FC in construction applications.

Rapid chloride permeability

Figure 10 demonstrates the effect of TAS cement replacement on the total charge passed by chloride ions in different FC mixes. As the percentage of TAS replacement increases from 0 to 40%, there is a marked reduction in the total charge passed, signifying improved resistance to chloride ion penetration. For example, the total

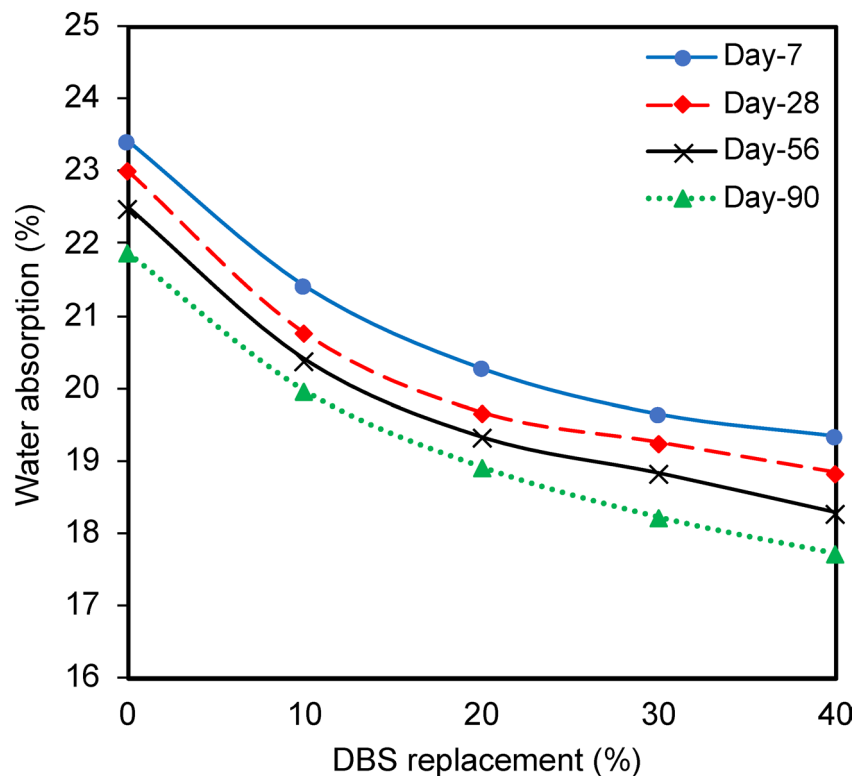


Fig. 8. Influence of TAS cement replacement on water absorption of different mixes.

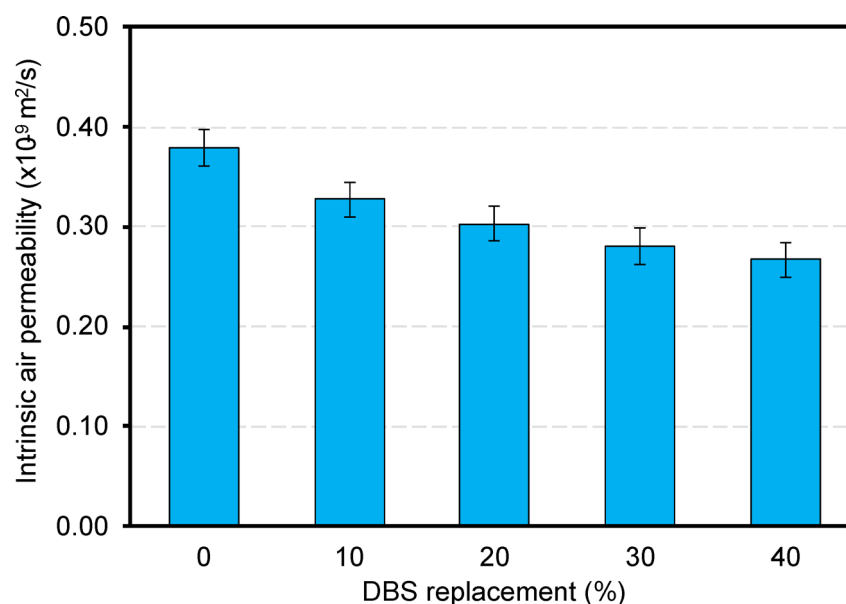


Fig. 9. Influence of TAS cement replacement on intrinsic air permeability of different mixes.

charge decreases from about 4864 coulombs for the control mix (0% TAS) to roughly 4607 coulombs at 40% TAS replacement. This indicates that a higher TAS content enhances the concrete's microstructure, making it denser and less permeable to chloride ions. Namsone et al.⁵³ studied the durability properties of high-performance FC and found that using supplementary cementitious materials (SCMs) can significantly enhance durability by reducing porosity and improving resistance to chloride ion penetration. Additionally, Chung et al.⁵⁴ investigated the microstructure of foamed cement paste with different SCMs and reported that these materials improve the mechanical properties and durability by enhancing the formation of C-S-H phases. Figure 11 shows the effect of TAS cement replacement on the chloride ion diffusion coefficient in FC. As the TAS replacement percentage

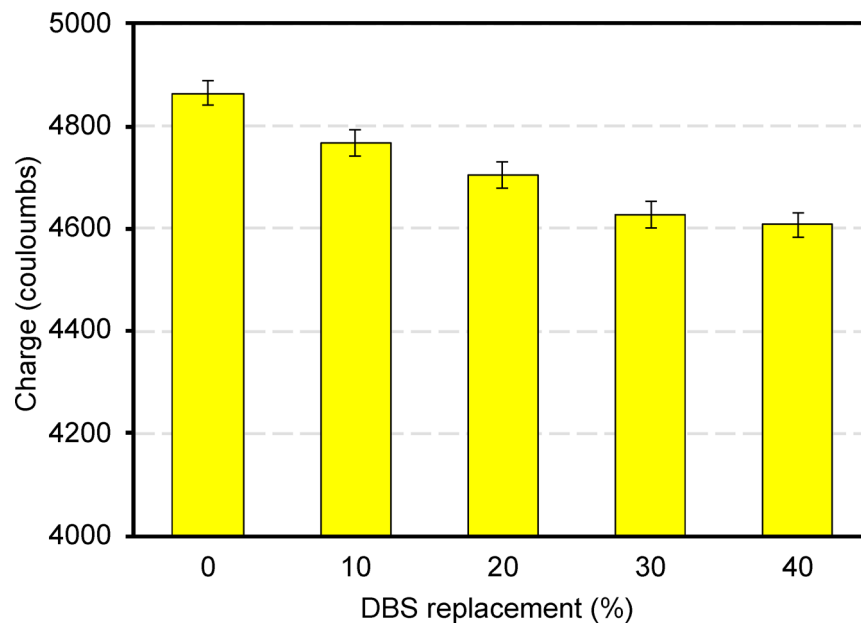


Fig. 10. Influence of TAS cement replacement on total charge of chloride ions of different mixes.

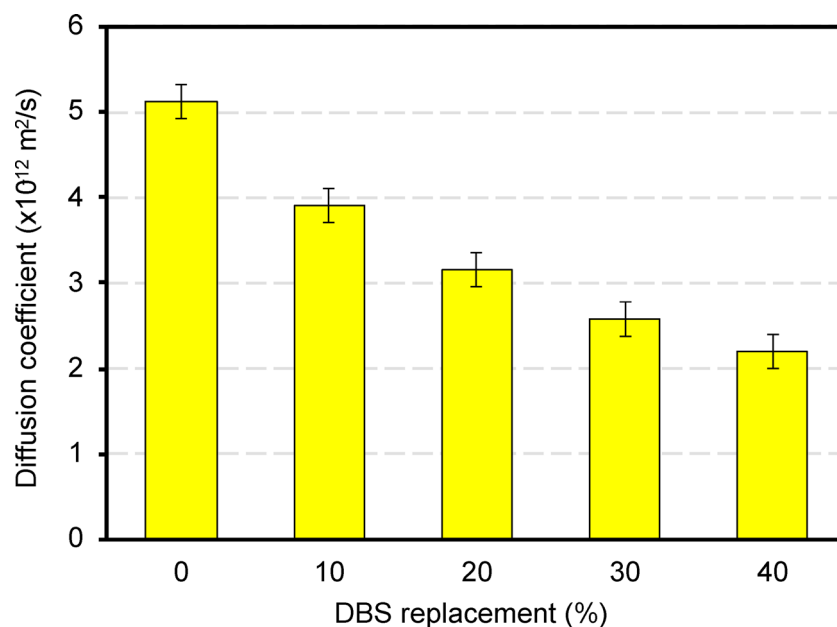


Fig. 11. Influence of TAS cement replacement on chloride ion diffusion coefficient of different mixes.

increases from 0 to 40%, the diffusion coefficient significantly decreases, from approximately $5.13 \times 10^{-12} \text{ m}^2/\text{s}$ for the control mix to about $2.21 \times 10^{-12} \text{ m}^2/\text{s}$ for at 40% TAS replacement. This reduction suggests that higher TAS content enhances the concrete's resistance to chloride ion penetration by improving the microstructure and decreasing porosity. Pizoń⁵² stresses the importance of accurately determining chloride diffusion coefficients to predict concrete performance. These studies align with the observed improvements in Fig. 11, showing that TAS effectively increases the durability of FC by reducing chloride ion permeability.

Mechanical properties

Compressive strength

Figure 12 demonstrates the impact of substituting TAS for cement on the compressive strength of FC over various curing periods (7, 28, 56, and 90 days). The results indicate that as the percentage of TAS replacement increases from 0 to 20%, compressive strength significantly improves, reaching a peak at 20% TAS replacement with 4.70 MPa at 90 days. However, further increasing the TAS content to 30% and 40% leads to a decline in

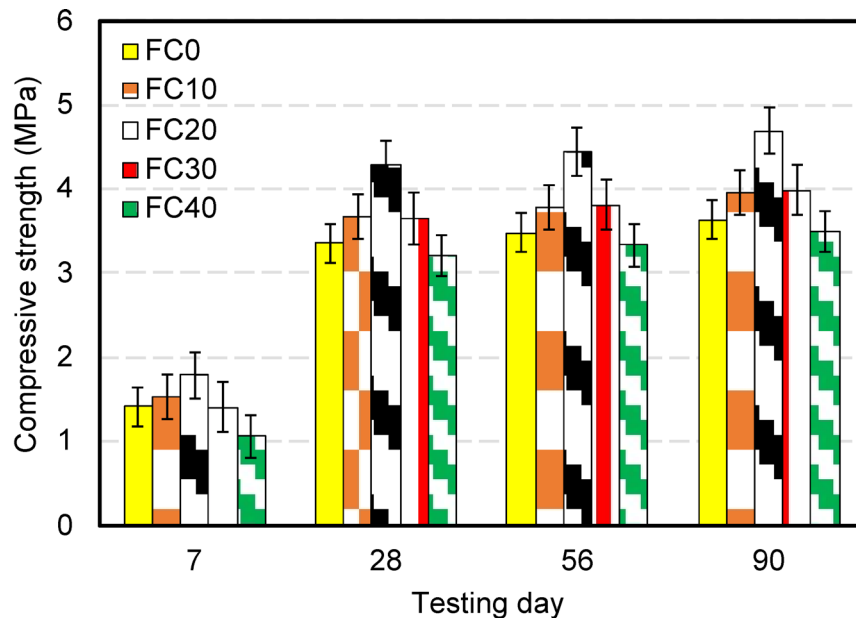


Fig. 12. Influence of TAS cement replacement on compressive strength of different mixes.

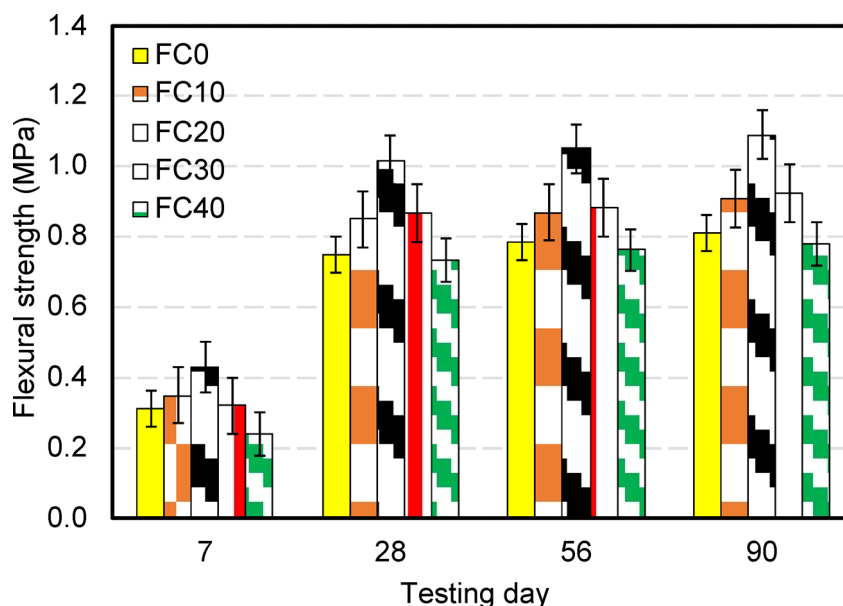


Fig. 13. Influence of TAS cement replacement on flexural strength of different mixes.

compressive strength, suggesting an optimal replacement level around 20%. This indicates that while TAS can enhance the compressive strength of FC up to a certain point, excessive replacement may negatively affect the concrete's microstructure. The compressive strength of FC increased with higher replacement percentages when processed spent bleaching earth was used as a partial cement replacement⁵². Similarly, the study conducted by Song et al.⁵⁵ discovered that the addition of lithium silicate enhanced the compressive strength and pore structure of FC, which is consistent with the benefits obtained as a result of TAS. These findings highlight the potential of TAS and comparable materials to improve the mechanical properties of FC when optimally employed.

Flexural strength

Figure 13 illustrates the effect of substituting TAS for cement on the flexural strength of various FC mixes at different curing periods (7, 28, 56, and 90 days). The data show that as the TAS replacement percentage increases from 0 to 20%, there is a significant rise in flexural strength. For example, at 28 days, the flexural strength increases from 0.75 MPa for the control mix (FC0) to 1.02 MPa for the mix with 20% TAS replacement (FC20).

However, further increasing the TAS content to 30% and 40% leads to a decrease in flexural strength, indicating that the optimal TAS replacement level is 20%. Chung et al.⁵⁴ confirm that incorporating additives can enhance the mechanical properties of foam concrete, including flexural strength, up to an optimal level before declining with higher replacement percentages. Additionally, Eltayeb et al.⁵⁶ found that using crumb rubber as a partial replacement for sand in foam concrete improved flexural strength up to a certain percentage, which is consistent with the observed improvements due to TAS. These findings highlight the potential of TAS to enhance the flexural strength of FC when used at optimal levels.

Split tensile strength

The influence of TAS cement replacement on the splitting tensile strength of various FC mixes at different curing periods (7, 28, 56, and 90 days) is illustrated in Fig. 14. The results indicate that increasing TAS replacement up to 20% enhances the splitting tensile strength, with the highest strength observed at 20% TAS replacement. For example, at 28 days, the tensile strength increases from 0.50 MPa for the control mix (FC0) to 0.66 MPa for the 20% TAS mix (FC20). However, further increasing the TAS content to 30% and 40% results in a decline in tensile strength, indicating an optimal replacement level at 20%. Research by Karrech et al.²⁴ supports these findings, demonstrating that the incorporation of TAS can enhance the mechanical properties of construction materials, including tensile strength, up to an optimal level before declining at higher replacement percentages. Additionally, the study by Fosu et al.⁵⁷ on the physicochemical characteristics of Tianqi aluminosilicate supports the notion that controlled incorporation, and processing can significantly improve mechanical properties such as tensile strength. These findings align with the observed trends for TAS, underscoring its potential to enhance the tensile strength of FC when used optimally.

Elastic modulus

Figure 15 illustrates the effect of TAS cement replacement on the elastic modulus of various mixes. Initially, there is an increase, followed by a decline. At 0% replacement, the elastic modulus is 1.130 GPa, which rises to 1.453 GPa at 20% replacement and then decreases to 1.012 GPa at 40% replacement. This pattern suggests that a moderate amount of TAS replacement improves the stiffness of concrete, likely due to enhanced microstructural properties, as observed in similar studies on supplementary cementitious materials¹⁴. However, excessive replacement (30% and above) may weaken the concrete matrix, leading to reduced mechanical performance, consistent with findings on alternative material replacements in concrete²⁰. In summary, the results show that optimal TAS replacement levels can improve the elastic modulus, while too much replacement may cause a decline in mechanical properties. This highlights the need to balance replacement percentages to maintain the structural integrity of concrete while enhancing its performance.

The reported pattern of an initial increase followed by a decrease in the elastic modulus with rising TAS concentration can be ascribed to multiple variables. As the TAS content rises, the density of the FC increases due to the microstructural densification, leading to an elevated elastic modulus. The findings in Sect. “Hardened density” substantiate this, demonstrating that increased TAS concentration elevates the material's density, hence enhancing interfacial bonding between the paste and aggregate. Nevertheless, beyond a specific threshold, the integration of elevated TAS content results in introducing more macro pores within the matrix. This leads to a decrease in the material's overall stiffness, resulting in a decline in the elastic modulus. The fluctuation in the modulus can be perceived as a combined effect of density, porosity, and the interfacial adhesion among the components. Moreover, the interfacial interaction between the TAS particles and the cement matrix is crucial

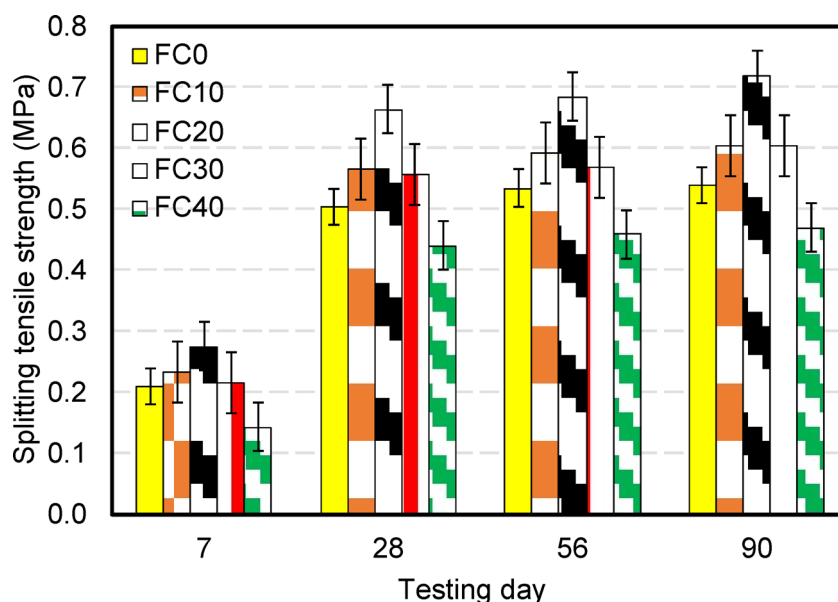


Fig. 14. Influence of TAS cement replacement on splitting tensile strength of different mixes.

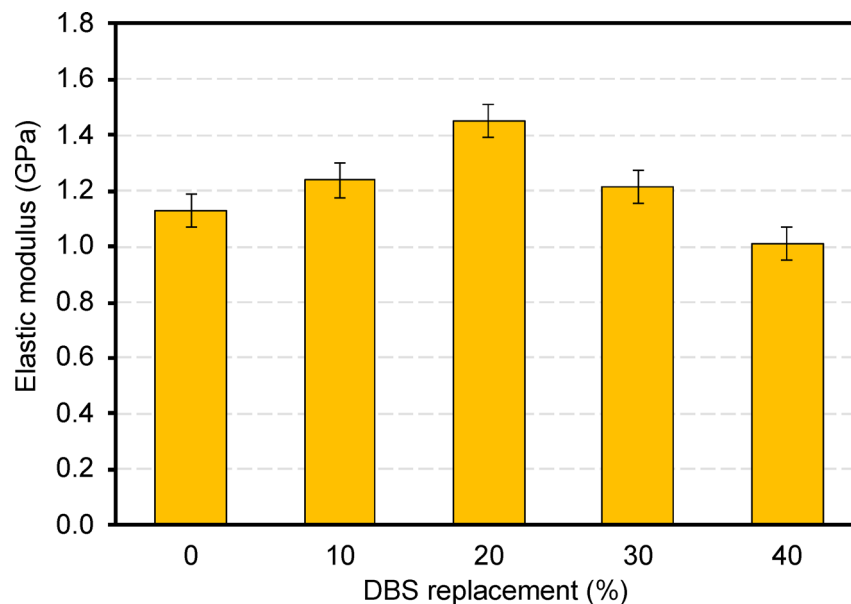


Fig. 15. Influence of TAS cement replacement on elastic modulus of different mixes.

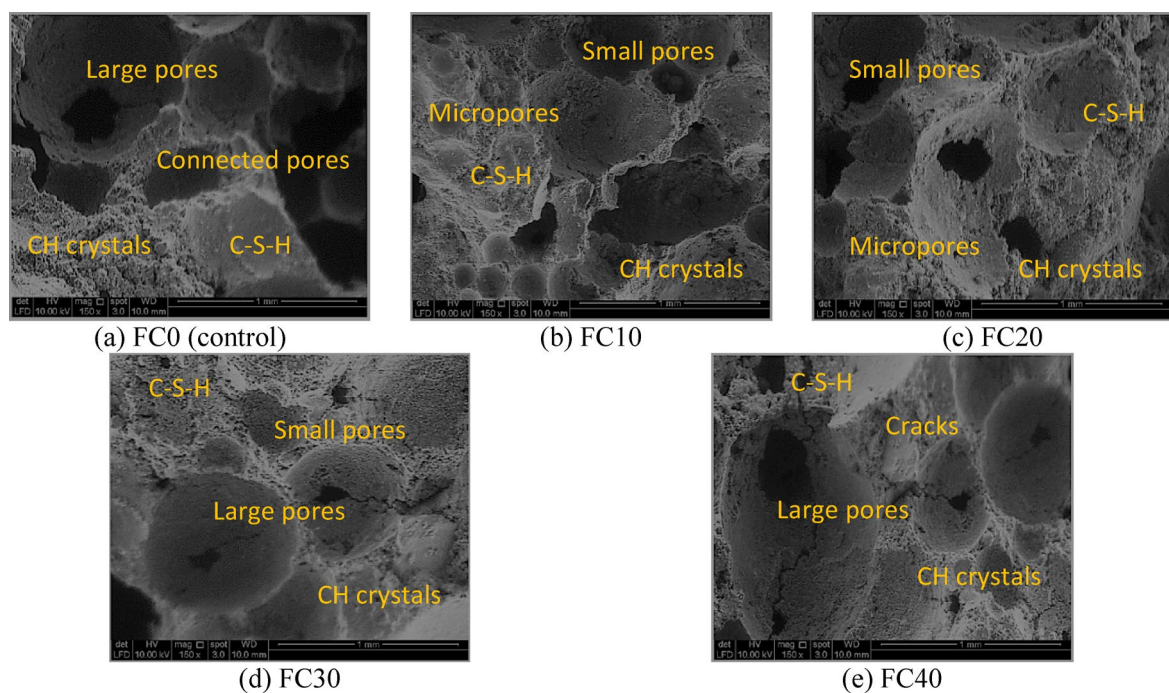


Fig. 16. SEM images of FC mixed with raw pumice.

in influencing the elastic modulus. The incorporation of moderate levels of TAS boosts its pozzolanic activity, hence enhancing the link between the matrix and aggregate and increasing the material's stiffness. Excessive TAS concentration may result in a diminished bond efficacy, particularly if the hydration products are inadequate to completely react with the available calcium hydroxide ($\text{Ca}(\text{OH})_2$). This leads to diminished strength and a decreased elastic modulus, seen at elevated replacement levels.

SEM assessment

The SEM images in Fig. 16 illustrate the microstructures of FC samples with varying percentages of TAS cement replacement: FC0 (control), FC10 (10% TAS), FC20 (20% TAS), FC30 (30% TAS), and FC40 (40% TAS). These images reveal significant variations in microstructural properties as the TAS content increases. The control sample (FC0) exhibits large and connected pores, CH (calcium hydroxide) crystals, and C-S-H gel. The large

pore characteristic of FC contributes to its lightweight nature but compromising its mechanical strength and durability due to increased permeability. The presence of CH crystals indicates ongoing hydration processes, contributing to overall strength but also suggesting potential porosity issues. The C–S–H gel, the primary binding phase, is present but not uniformly distributed, resulting in a less dense and more porous structure.

As TAS replacement increases to 10% (FC10), significant microstructural improvements are observed. The SEM image shows smaller pores and micropores, CH crystals, C–S–H gel, and a denser phase, suggesting that 10% TAS helps fill larger voids, resulting in a denser concrete matrix. The well-distributed C–S–H gel indicates improved hydration and binding properties, crucial for concrete strength and durability. The FC20 sample exhibits further improvements, with a dense phase, small pores, and well-distributed C–S–H gel. Enhanced packing and better material compaction contribute to increased mechanical properties, aligning with research indicating that TAS optimizes FC by improving its microstructure and reducing porosity²⁴. Supporting studies, such as those on palm oil fuel ash (POFA), show that similar additives enhance concrete properties by filling voids and improving hydration, leading to a denser and more durable matrix⁵⁸.

However, at higher TAS replacement levels, such as 30% (FC30) and 40% (FC40), the SEM images reveal a decline in microstructural quality. The FC30 sample shows larger pores and increased connectivity, suggesting reduced matrix compactness and diminished mechanical properties. The FC40 sample highlights significant microstructural degradation, with large pores, cracks, and less uniformly distributed C–S–H gel, resulting in substantial porosity and reduced material integrity. These features align with findings that excessive TAS content can lead to increased porosity and reduced strength²⁴. Research indicates that overuse of certain cement replacements, such as POFA, can lead to microstructural degradation and decreased mechanical properties, emphasizing the need to maintain optimal TAS replacement levels for enhanced durability in FC⁵⁸.

Pore size distribution

Figure 17 illustrates that the incorporation of TAS into FC mixtures resulted in an alteration in the arrangement of pore sizes, characterised by an increase in the number of voids with diameters of 400 µm or less and a decrease in those with diameters above 400 µm. The amount of micro-pores with widths less than 400 µm significantly rose upon the introduction of TAS at concentrations up to 20%. The particle size of TAS employed in FC has been shown to be adequate for pore bridging. Nevertheless, increasing the quantity of TAS integrated into the FC augments both pore dimensions and distribution. In comparison to the control specimen (F0), TAS-based FC exhibited a higher pore frequency of less than 400 µm. The inclusion of TAS markedly reduced macro-pores to a diameter of under 500 nm. The C–S–H gel diminished the total porosity by occupying the voids. The admixture particles, because of their remarkable micro-aggregate action and extensive specific surface area, occupied the pores and diminished the overall porosity. The degree of secondary hydration diminished, but total porosity augmented following the incorporation of 20% TAS into the C–S–H gel, which inhibited its capacity to occupy more nano-pores; these findings are consistent with several studies^{59–61}. The frequency was adversely impacted by an elevation in TAS percentage beyond 20%, resulting in a reduction of holes measuring 400 µm or less and an augmentation of pores ranging from 400 to 1000 nm or greater.

Conclusion

1. Incorporating Tianqi aluminosilicate (TAS) as a cement replacement in foamed concrete (FC) significantly improves its mechanical and transport properties. The optimal replacement level of TAS, around 20%, results in notable improvements in compressive strength, flexural strength, splitting tensile strength, and elastic modulus.
2. SEM analysis reveals that TAS enhances the concrete's microstructure by reducing pore size, promoting a better-distributed C–S–H gel, and minimizing the presence of large, interconnected pores. These changes contribute to the observed improvements in the concrete's strength and durability.
3. The inclusion of TAS reduces permeable porosity and intrinsic air permeability while enhancing resistance to chloride ion penetration. These improvements indicate that TAS contributes to better long-term durability and makes FC more resistant to environmental factors, such as water and chloride ingress.
4. While 20% replacement of cement with TAS improves performance, higher levels of TAS replacement may lead to diminished workability and strength. This presents challenges for large-scale applications where uniform material properties are critical for consistent performance.
5. Future studies should explore the effects of higher TAS content beyond 20%, investigate its combination with other supplementary cementitious materials (SCMs), assess the economic feasibility of large-scale TAS production, and conduct life cycle analysis (LCA) to evaluate its environmental impact, particularly regarding its carbon footprint and sustainability contributions.

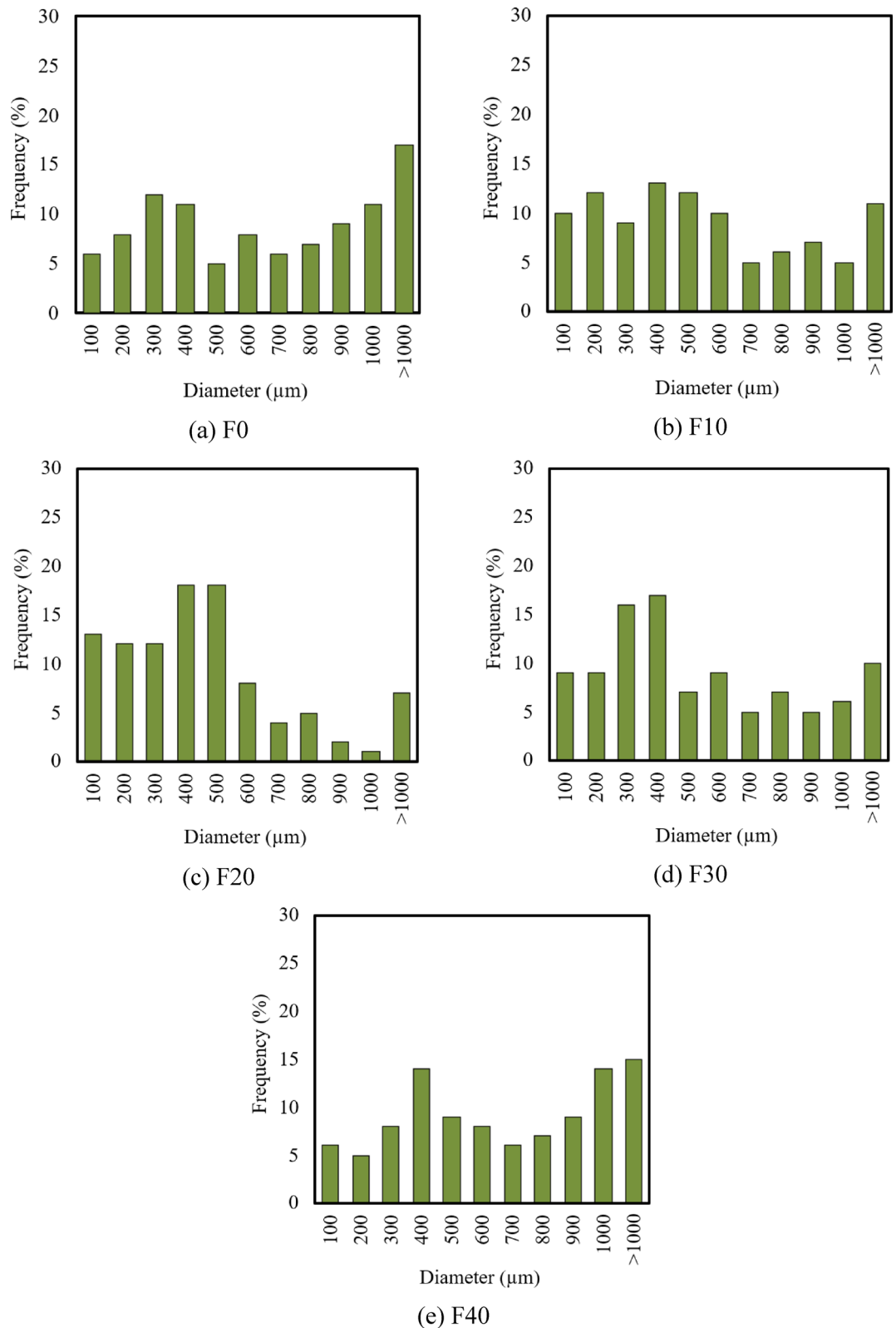


Fig. 17. Pore distribution of varying FC mixtures.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

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Author contributions

AAS—Investigation, analysis, writing and review, MAOM—Conceptualization, supervision, methodology, review and editing, AN—Methodology, investigation, Review and editing, MMAA—Methodology, investigation, Review and editing, POA—Conceptualization, supervision, methodology, writing original draft, Review and editing, OGF—Methodology, investigation, review and editing.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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